

CSE421 Assignment 05

[MSMA | Summer 2025]

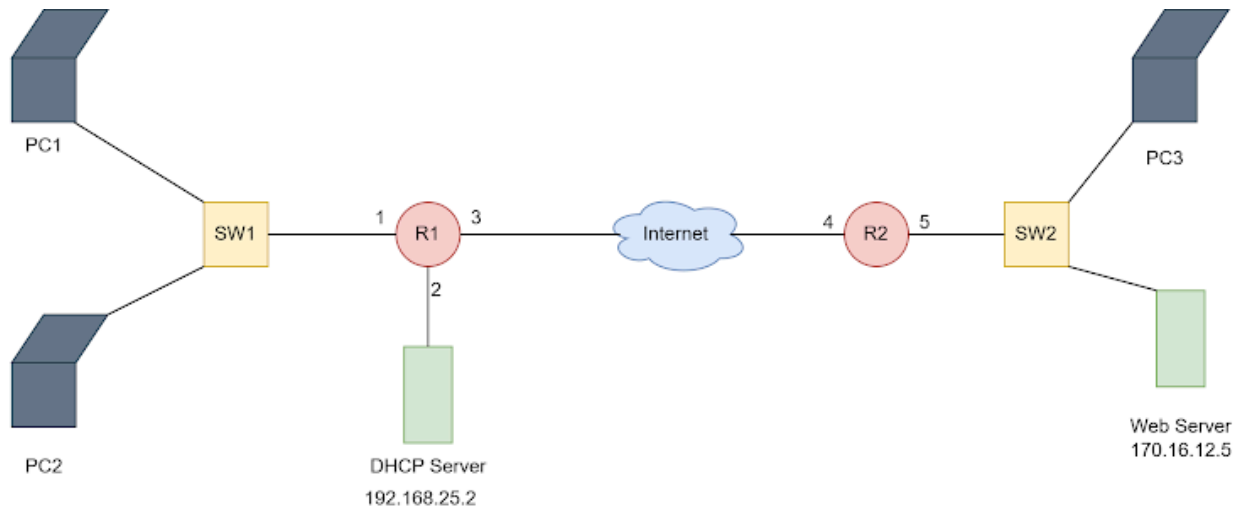
Answer all the questions in this form. You can submit only once even if you submit by mistake. Make sure that your answers are put correctly by refreshing the page.

Q1. Which options of the following are correct?

1. Interfaces of a router can never act as both input port and output port
2. Switching fabric does routing
3. Internet follows Datagram network
4. Switching is decentralized

Answer: 3, 4

Answer the rest of the questions based on the following topology.



Q2. Both PC1 and PC2 are part of the 192.168.9.0/24 network and they will get their IP address with the help of DHCP. Select the correct option.

1. No DHCP relay needed for the PCs to get IP configuration

2. DHCP relay is needed to be set in port 1
3. DHCP relay is needed to be set in port 2
4. DHCP relay is needed to be set in port 3

Answer: 2

Q3. What will be the destination IP address when the DHCP Discover packet is going from PC1 to R1? (Format: 10.10.10.10)

Answer: 255.255.255.255

Q4. What will be the destination IP address when the DHCP Discover packet is going from R1 to DHCP Server? (Format: 10.10.10.10)

Answer: 192.168.25.2

Q5. If both PC1 and PC2 want to go to the internet at the same time how will that be done if R1 has only one public IP address? (Not more than two words)

Answer: PAT

Q6. The web server doesn't have any dedicated public IP address. How will the devices outside of its network request the server? (Not more than two words)

Answer: Port Forwarding

Answer Q7 to Q10 based on the following scenario and the above topology

PC1 got IP address **192.168.9.20** and PC2 got IP address **192.168.9.10** through DHCP. Both R1 and R2 are the border gateway routers and have only one public IP address each for going to the internet. R1 has **209.20.20.20** in its NAT pool and R2 has **209.10.10.10** in its NAT pool.

PC1 has requested a webpage from the webserver. The source port address is **62000** and the destination port address is **80**. Immediately, PC2 also requested for a webpage from the webserver using the port address **62000**.

Q7. What will be the source socket address when the packet is going from PC2 to R1? (Format: 10.10.10.10:100)

Answer: 192.168.9.10:62000

Q8. What will be the source socket address when the packet is going from R1 to Webserver? (Format: 10.10.10.10:100)

Answer: 209.20.20.20:62001

Q9. What will be the destination socket address when the packet is going from PC2 to R2? (Format: 10.10.10.10:100)

Answer: 209.10.10.10:80

Q10. What will be the destination socket address when the packet is going from R2 to Webserver? (Format: 10.10.10.10:100)

Answer: 170.16.12.5:80

Answer Q11 to Q15 based on the following scenario and the above topology

The request packet R1 got from PC1 is X bytes. However, the MTU of the link between R1 and the Internet is less than X bytes. So, the packet got fragmented into **29 fragments** and one of them is **one-fourth of MTU**. The last byte number of the **12th fragment** is **13703** and the fragment offset of the **18th fragment** is **2343**. The header size is **one-eighth of the MTU**.

Hint: Starting byte number is not 0.

Q11. Calculate the size of MTU.

Answer: 1152

$$(2343*8-13704)/5=1008$$

$$\text{let, mtu} = x, \text{ header} = x/8$$

$$1008 + x/8 = x$$

$$x = 1152$$

Q12. Calculate the size of the original packet.

Answer: 28512

$$\text{header} = 144 \text{ one eighth of mtu}$$

$$\text{data in a full fragment} = 1008$$

$$\text{last fragment size} = 1152/4=288$$

$$\text{original packet size} = 1008*28+(288-144) + 144=28512$$

Q13. Calculate the starting byte number of data.

Answer: 1608

$$\text{starting} = 13704-12*1008=1608$$

Q14. Calculate the fragment offset of the 26th fragment.

Answer: 3351

$$(1608+1008*25)/8=3351$$

Q15. If all the fragments arrive at R2 and R2 needs to pass the fragments using a link having MTU of 424 bytes, calculate the number of fragments that will be created in total for the original datagram.

Answer: 113

$$\text{new mtu} = 424$$

max data in new fragments = $424 - 144 = 280$

data in previous full fragments = 1008

one full fragment will be divided into $1008 / 280 = 4$ (ceil value)

There were 28 full fragments, they will be divided into 112 fragments.

Now, the last fragment that came of R1 had data = $288 - 144 = 144$

For this we need $144 / 280 = 1$ fragment.

Total number of fragments R2 will create = $112 + 1 = 113$